(1, 25, 31, 64, 79, 102)

		64	32	16	8	4	2	1
	0	0	0	0	0	0	0	1
25 →	0	0	0	1	1	0	0	1
31 →	0	0	0	1	1	1	1	1
64 →	0	1	0	0	0	0	0	0
79 →	0	1	0	0	1	1	1	1
102 →	0	1	1	0	0	1	1	0

96

Binary coded decima

1	0 0000001
64	1 0000000
25	00 11001
102	1 1001 10
31	00 11111
79	1001111

I

cannot pair

$\therefore f = \bar{a}\bar{b}\bar{c}\bar{d}\bar{e}\bar{f}g + a\bar{b}\bar{c}\bar{d}\bar{e}\bar{f}\bar{g} + \bar{a}\bar{b}cd\bar{e}$

$+ ab\bar{c}\bar{d}efg + \bar{a}\bar{b}cdefg + a\bar{b}\bar{c}defg$

Combinational circuit

Present i/p ↔ prest o/p

NO memory

NO feedback

NO clock

Sequential

past O/P + P.I → P.O.

Memory (o/p)

Feedback

clock

Code converter : comb. circuit takes i/p in one format & converts into equivalent other format.

BCD to Excess-3 ; BCD to gray code equivalent.

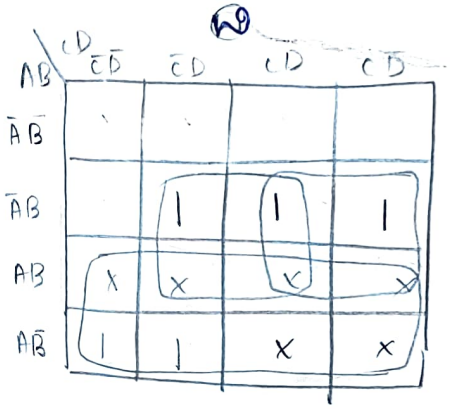
BCD i/p	Excess-3		
0 0000	0011	6 0101	1000
1 0001	0100	7 0111	1001
2 0010	0101	8 1000	1011
3 0011	0110	9 1001	1100
4 0100	0111		

only upto 9

code conversion from BCD to Excess-3

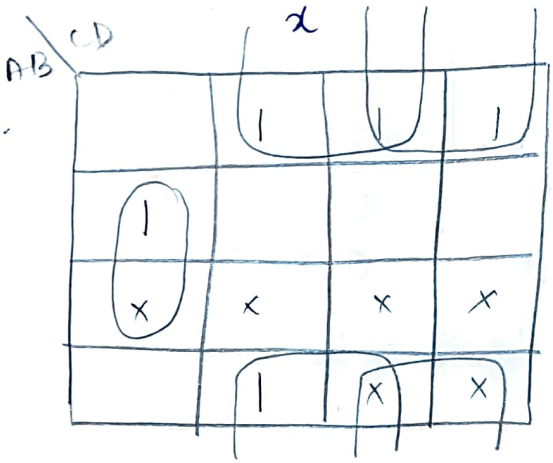
A	B	C	D	W	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

W



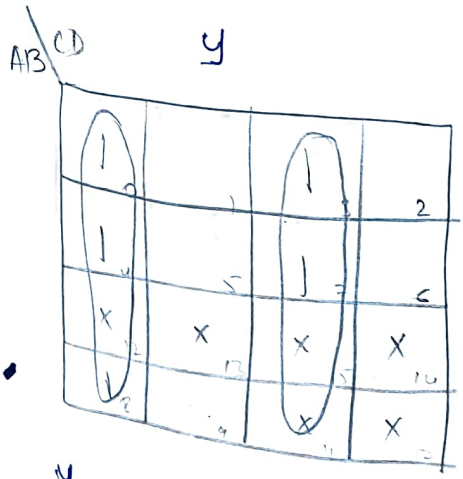
$W = A + BD + B\bar{C}$

x



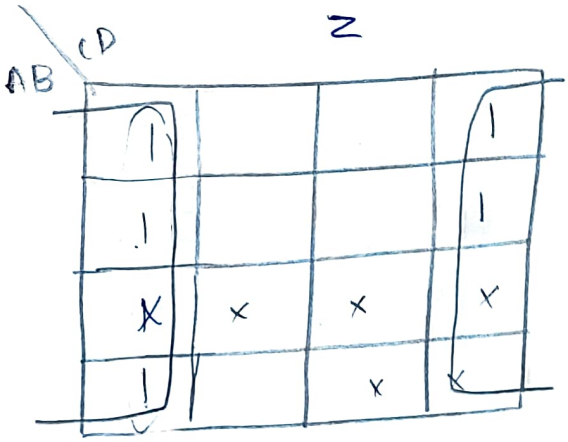
$x = \bar{B}D + \bar{B}C + B\bar{C}\bar{D}$

y



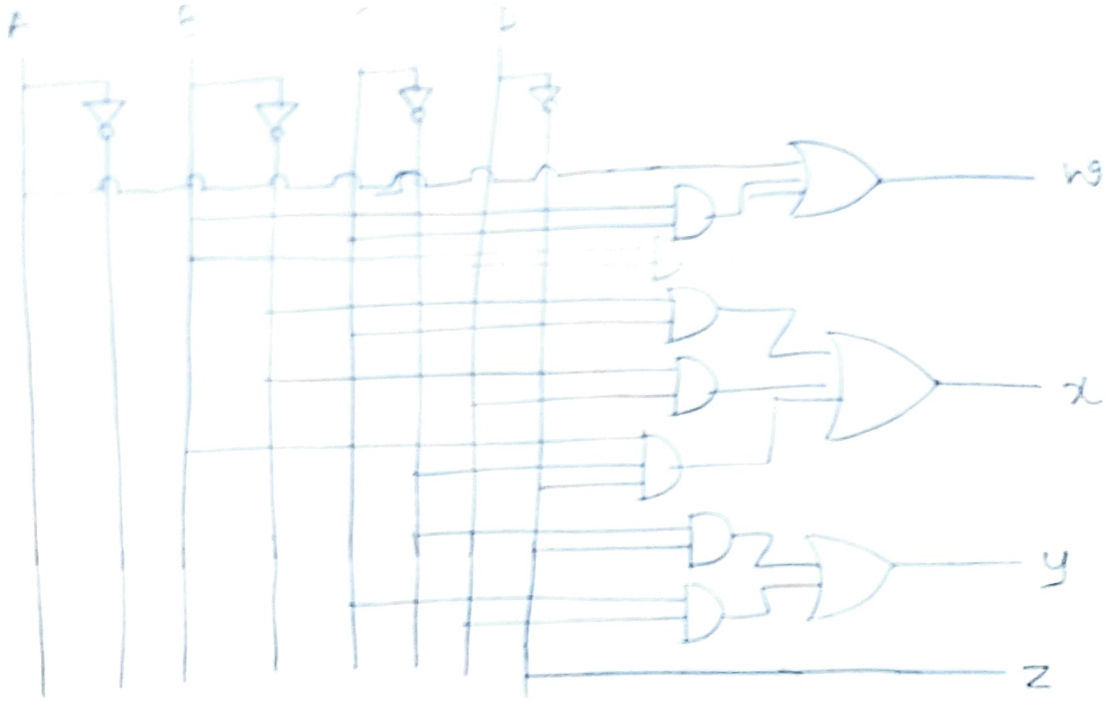
$y = \bar{C}\bar{D} + CD = \overline{C \oplus D}$

z



$z = \bar{D}$

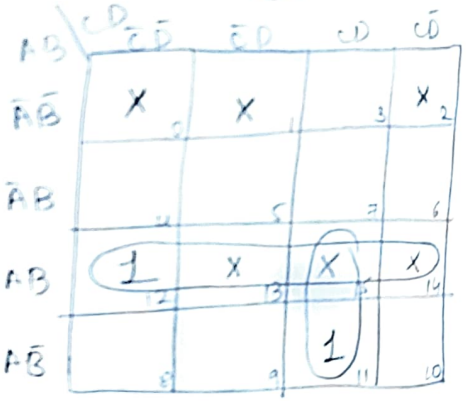
Logic representation



Code conversion from excess-3 to BCD

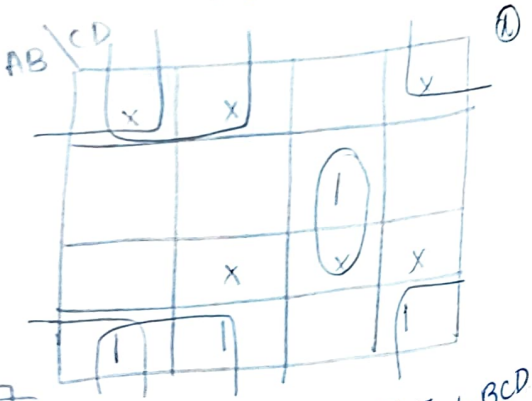
A	B	C	D	w	x	y	z
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1

(w)

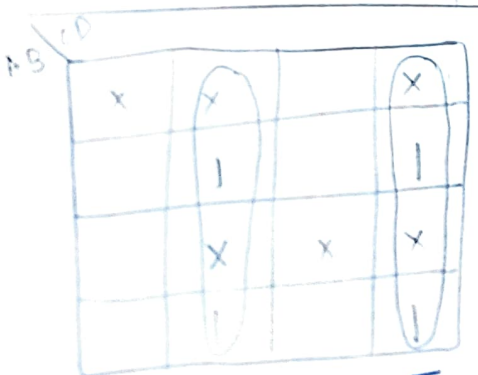


$W = AB + ACD$

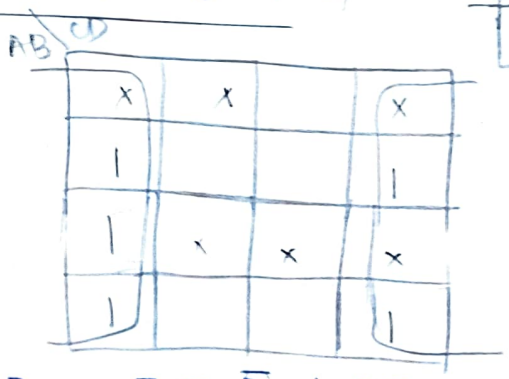
(x)



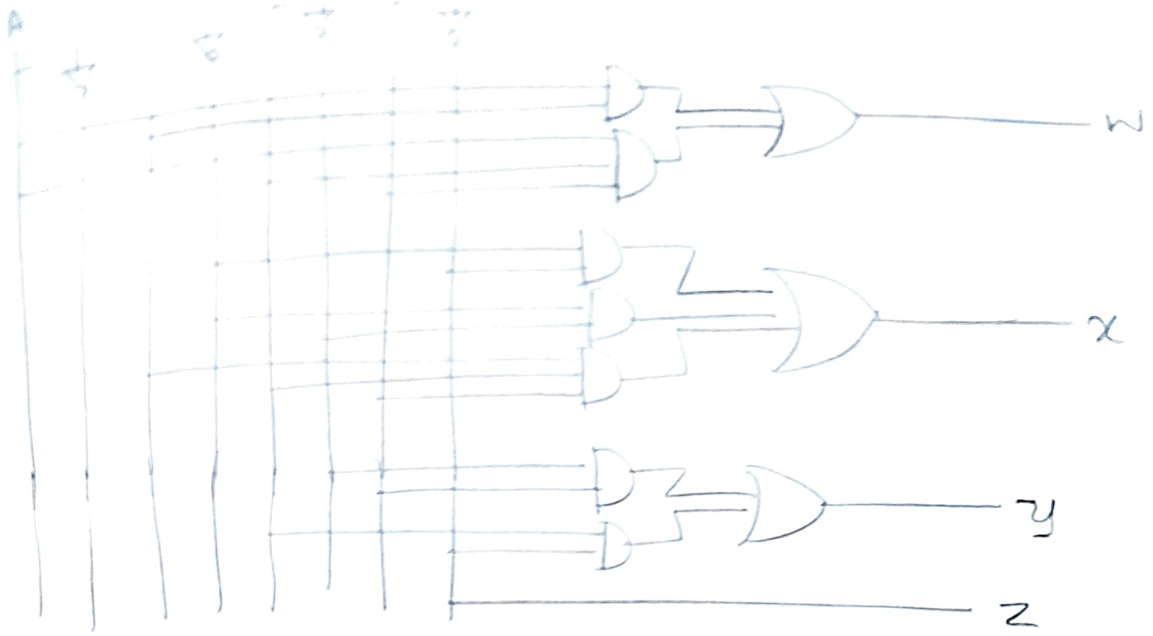
$x = \bar{B}\bar{D} + \bar{B}\bar{C} + BCD$



$y = \bar{C}D + C\bar{D} = C \oplus D$

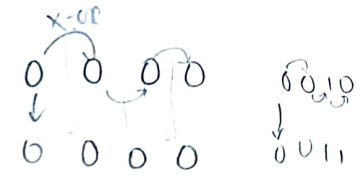


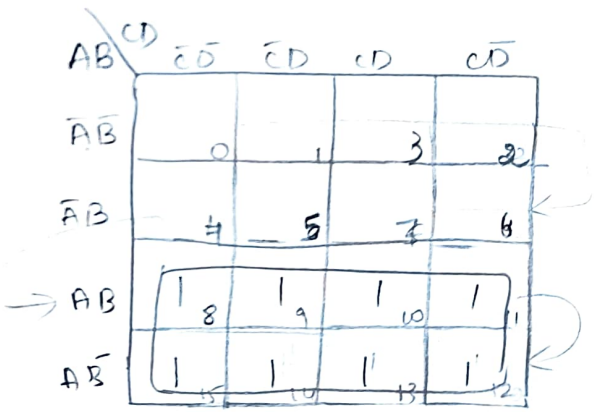
$z = \bar{D}$ ~~$+ A\bar{A}B$~~



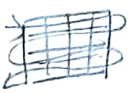
Binary code to gray code conversion:

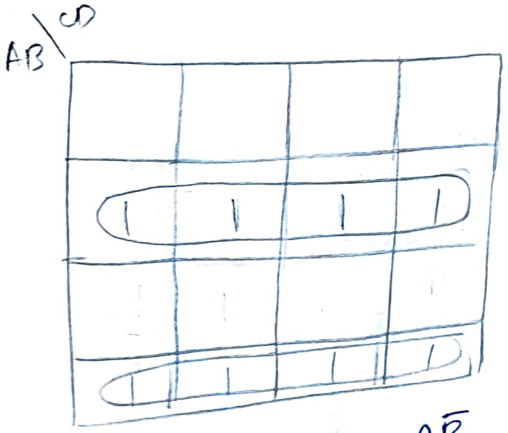
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1
10	1	0	1	0	1	1	1	1
11	1	0	1	1	1	1	1	0
12	1	1	0	0	1	0	1	0
13	1	1	0	1	1	0	1	1
14	1	1	1	0	1	0	0	1
15	1	1	1	1	1	0	0	0





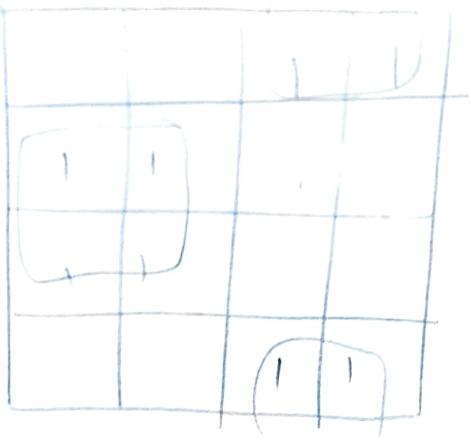
$W = A$



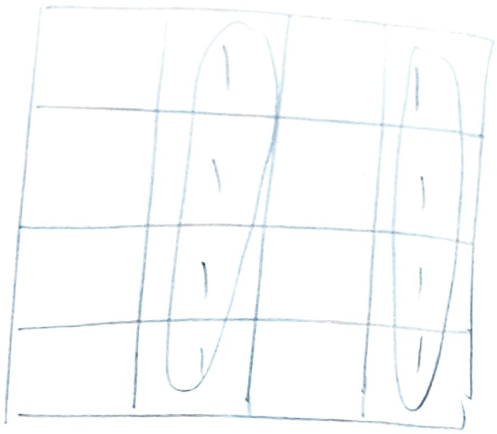


$X = \bar{A}B + A\bar{B}$

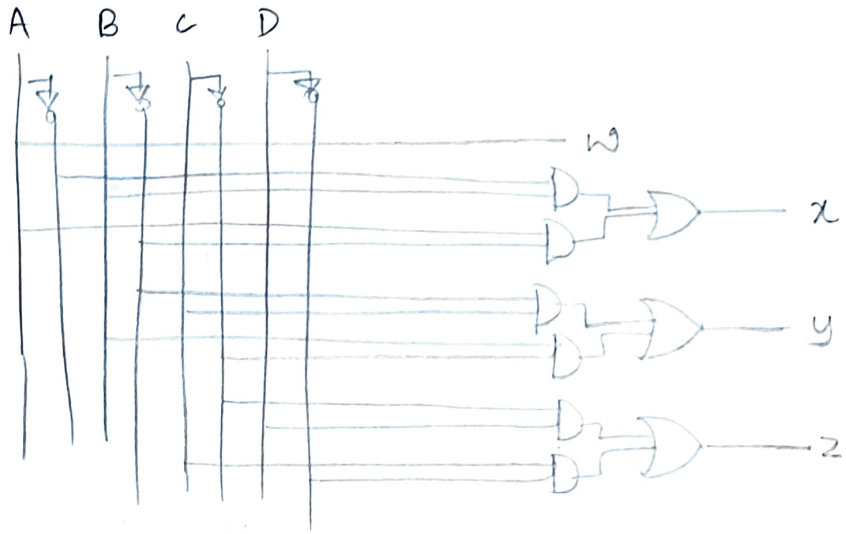
$= A \oplus B$



$y = \bar{b}c + b\bar{c}$



$z = \bar{c}D + c\bar{D}$



Code format

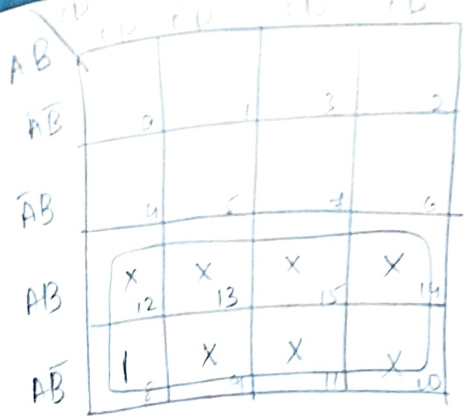
	8 (A)	−4 (B)	−2 (C)	−1 (D)
0	0	0	0	0
1	1	1	1	1
2	1	1	0	0
3	1	1	0	1
4	1	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0

8 4 2 1 BCD

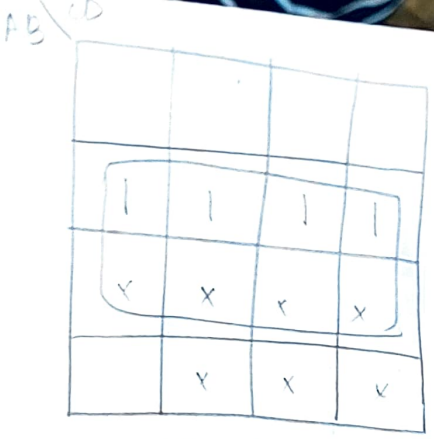
	w	x	y	z
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0

not standard format so not used

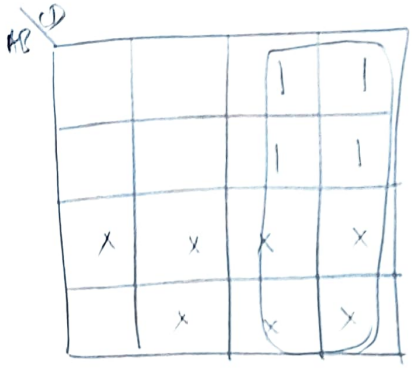
A — w
B — x
C — y
D — z



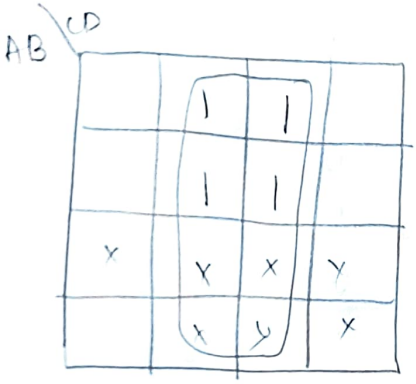
$w = A$



$x = B$



$y = C$

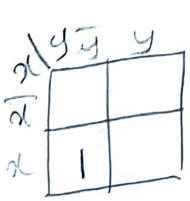


$z = D$

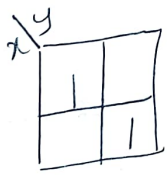
Comparators: is a combinational circuit which compares the magnitude of binary numbers and gives whether one no. is equal to other no., less than the other no., greater than the other number.

1 bit comparator:

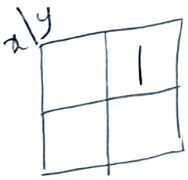
x	y	$x > y$	$x = y$	$x < y$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0



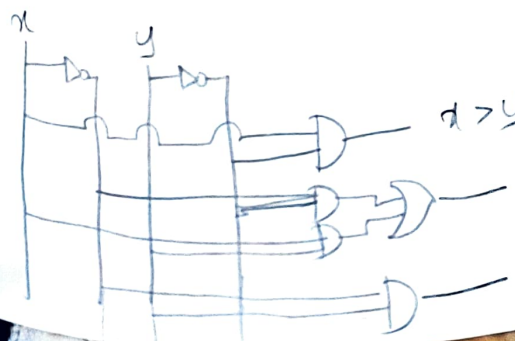
$x > y = x\bar{y}$



$x = y = \bar{x}\bar{y} + xy$
$= \overline{x \oplus y}$

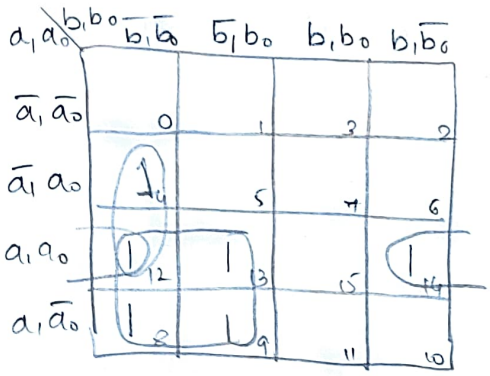


$x < y = \bar{x}y$

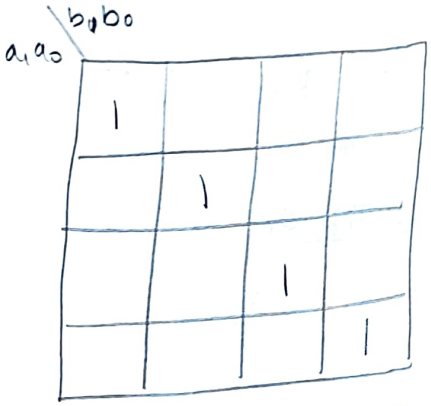


Design and implement 2-bit comparator using logic gates:

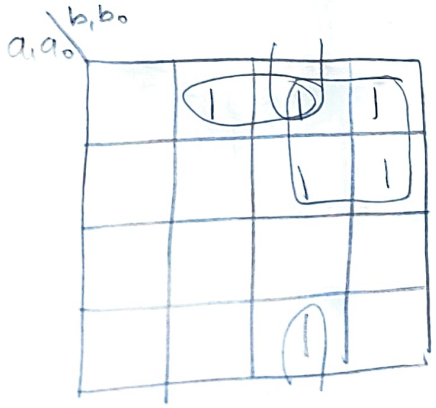
	a_1	a_0	b_1	b_0	$a_1a_0 > b_1b_0$	$a_1a_0 = b_1b_0$	$a_1a_0 < b_1b_0$
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1
4	0	1	0	0	1	0	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	0	1
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0



$$a_1a_0 > b_1b_0 = a_1\bar{b}_1 + a_0\bar{b}_1\bar{b}_0 + a_1a_0\bar{b}_0$$



$$a_1a_0 = b_1b_0 = \bar{a}_1\bar{a}_0\bar{b}_1\bar{b}_0 + a_1a_0b_1b_0 + a_1\bar{a}_0b_1\bar{b}_0$$



$$a_1a_0 < b_1b_0 = \bar{a}_1b_1 + \bar{a}_1\bar{a}_0b_0 + \bar{a}_0b_1b_0$$

implement using logic gates ✱

Represent the truth table of 3 bit comparator

	a_2	a_1	a_0	b_2	b_1	b_0	$a_2a_1a_0 > b_2b_1b_0$	$a_2a_1a_0 = b_2b_1b_0$	$a_2a_1a_0 < b_2b_1b_0$
0	0	0	0	0	0	0	0	1	0
1	0	0	0	0	0	1	0	0	1
2	0	0	0	0	1	0	0	0	1
3	0	0	0	0	1	1	0	0	1
4	0	0	0	1	0	0	0	0	1
5	0	0	0	1	0	1	0	0	1
6	0	0	0	1	1	0	0	0	1
7	0	0	0	1	1	1	0	0	1
8	0	0	1	0	0	0	1	0	0
9	0	0	1	0	0	1	0	1	0
10	0	0	1	0	1	0	0	0	1
11	0	0	1	0	1	1	0	0	1
12	0	0	1	1	0	0	0	0	1
13	0	0	1	1	0	1	0	0	1
14	0	0	1	1	1	0	0	0	1
15	0	0	1	1	1	1	0	0	1
16	0	1	0	0	0	0	1	0	0
17	0	1	0	0	0	1	1	0	0
18	0	1	0	0	1	0	0	1	0
19	0	1	0	0	1	1	0	0	1
20	0	1	0	1	0	0	0	0	1
21	0	1	0	1	0	1	0	0	1
22	0	1	0	1	1	0	0	0	1
23	0	1	0	1	1	1	0	0	1
24	0	1	1	0	0	0	1	0	0
25	0	1	1	0	0	1	1	0	0
26	0	1	1	0	1	0	1	0	0

29	0	1	1	1	1	0	0	0	0
30	0	1	1	1	1	0	1	0	0
31	0	1	1	0	0	0	1	0	0
32	1	0	0	0	0	1	1	0	0
33	1	0	0	0	1	0	1	0	0
34	1	0	0	0	1	1	1	0	0
35	1	0	0	0	1	0	0	1	0
36	1	0	0	0	0	0	1	0	0
37	1	0	0	1	0	1	0	0	0
38	1	0	0	1	1	0	0	0	1
39	1	0	0	1	1	1	1	0	0
40	1	0	1	0	0	0	1	0	0
41	1	0	1	0	0	1	1	0	0
42	1	0	1	0	1	0	1	0	0
43	1	0	1	0	1	1	1	0	0
44	1	0	1	1	0	0	1	0	0
45	1	0	1	1	0	1	0	1	0
46	1	0	1	1	1	0	0	0	1
47	1	0	1	1	1	1	0	0	0
48	1	0	0	0	0	0	1	0	0
49	1	1	0	0	0	1	1	0	0
50	1	1	0	0	1	0	1	0	0
51	1	1	0	0	1	1	1	0	0
52	1	1	0	1	0	0	1	0	0
53	1	1	0	1	0	1	1	0	0
54	1	1	0	1	1	0	0	1	0

55	1	1	0	1	1	1	0	0	1
56	1	1	1	0	0	0	1	0	0
57	1	1	1	0	0	1	1	0	0
58	1	1	1	0	1	0	1	0	0
59	1	1	1	0	1	1	1	0	0
60	1	1	1	1	0	0	1	0	0
61	1	1	1	1	0	1	1	0	0
62	1	1	1	1	1	0	1	0	0
63	1	1	1	1	1	1	0	1	0

Parity bit generator & checker

Extra bit added to message to make the o/p to be odd or even

① odd parity ② Even parity

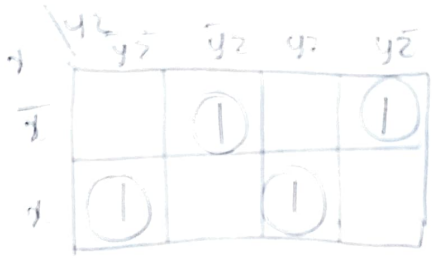
checker
0 → correct
1 → error

- Design and implement even parity generator and checker.

Message

x	y	z	P_{even}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

0 → to make odd
1 → " even



$$\bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$$

$$\bar{x}(\bar{y}z + y\bar{z}) + x(\bar{y}\bar{z} + yz)$$

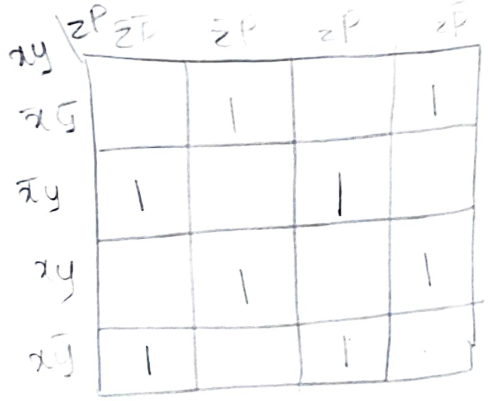
$$\bar{x}(y \oplus z) + x(\overline{y \oplus z})$$

$$x \oplus y \oplus z$$



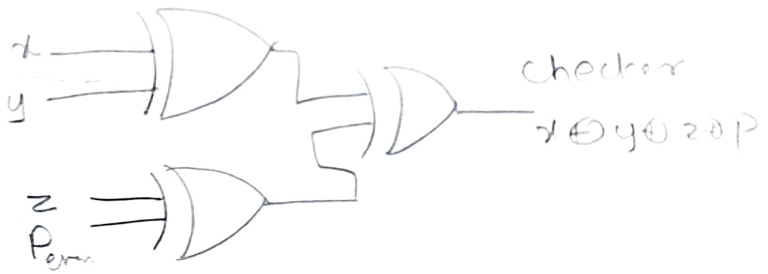
Even parity generator

x	y	z	P_{even}	$checker_{even}$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1



checker = $x \oplus y \oplus z \oplus P_{even}$

1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



X-OR

1. 4-bit adder / subtra
2. Parity generator & checker

• Design and implement odd parity and checker.

x	y	z	Podd
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

x \ yz	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1		1	
x		1		1

$\bar{x}\bar{y}\bar{z} + \bar{x}yz + x\bar{y}z + xy\bar{z}$

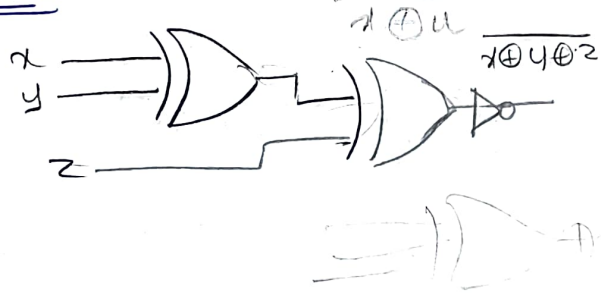
$\bar{x}(\bar{y}\bar{z} + yz) + x(\bar{y}z + y\bar{z})$

$\bar{x}(\overline{y \oplus z}) + x(y \oplus z)$ $\quad u = y \oplus z$

$= \overline{x \oplus y \oplus z}$

$x\bar{u} + xu$

$x \oplus u$



x	y	z	Podd	checker odd
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

xy \ zP	$\bar{z}\bar{P}$	$\bar{z}P$	$z\bar{P}$	zP
$\bar{x}\bar{y}$	1		1	
$\bar{x}y$		1		1
$x\bar{y}$	1		1	
xy		1		1

checker = $\bar{x}\bar{y}\bar{z}\bar{P} + \bar{x}\bar{y}zP + \bar{x}y\bar{z}P + \bar{x}yz\bar{P} + xy\bar{z}\bar{P} + xyzP + x\bar{y}\bar{z}P + x\bar{y}z\bar{P}$

$= \bar{x}\bar{y}(\bar{z}\bar{P} + zP) + \bar{x}y(\bar{z}P + z\bar{P}) + xy(\overline{z\oplus P}) + x\bar{y}(z\oplus P)$

$= \bar{x}\bar{y}(\overline{z\oplus P}) + \bar{x}y(z\oplus P) + xy(\overline{z\oplus P}) + x\bar{y}(z\oplus P)$

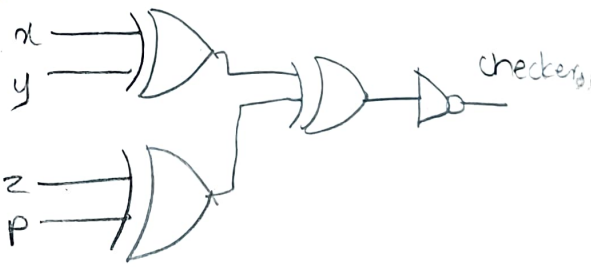
$= \bar{x}\bar{y}\bar{u} + \bar{x}yu + xy\bar{u} + x\bar{y}u$ let $u = z\oplus P$

$= \bar{x}(\bar{y}\bar{u} + yu) + x(y\bar{u} + \bar{y}u)$

$= \bar{x}(\overline{y\oplus u}) + x(y\oplus u)$

$= \overline{x\oplus y\oplus u}$

$= x\oplus y\oplus z\oplus P$



Multiplexer -> read

Implement $u(a,b,c,d) = ad + b\bar{c} + bd$ using 16:1, 8:1 and 4:1 MUX.

$u(a,b,c,d) = ad + b\bar{c} + bd$

$= a(b+\bar{b})(c+\bar{c})d + (a+\bar{a})b\bar{c}(d+\bar{d}) + (a+\bar{a})b(c+\bar{c})d$

$= (ab+a\bar{b})(cd+\bar{c}d) + (abc\bar{c} + \bar{a}b\bar{c})(d+\bar{d}) + (ab+\bar{a}b)(cd+\bar{c}d)$

$= abcd + ab\bar{c}d + a\bar{b}cd + a\bar{b}\bar{c}d + ab\bar{c}d + ab\bar{c}\bar{d} + \bar{a}b\bar{c}d + \bar{a}b\bar{c}\bar{d}$

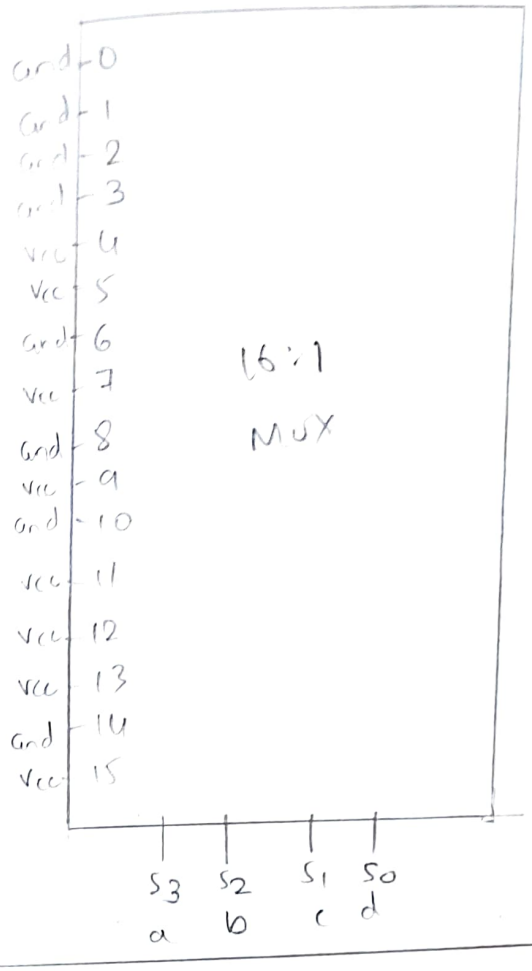
$abcd + ab\bar{c}d + \bar{a}bcd + \bar{a}b\bar{c}d$

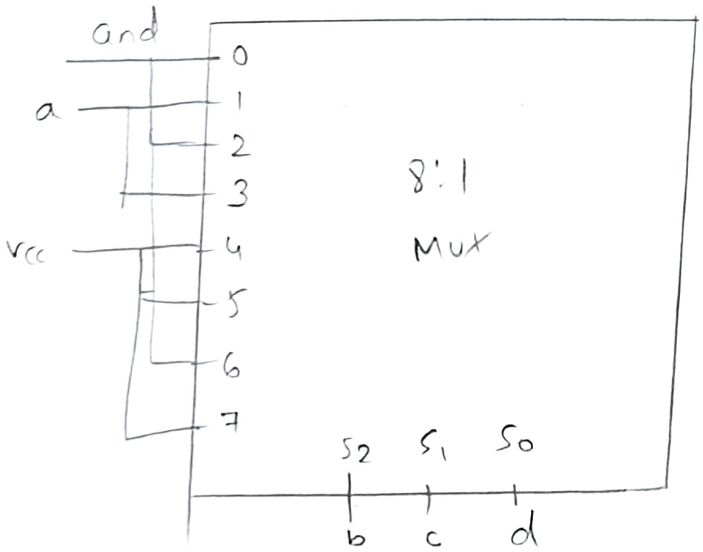
$= \Sigma m(4, 5, 7, 9, 11, 12, 13, 15)$

16:1 $16 = 2^{(4)}$ → selection lines

$\bar{a}$	0	1	2	3	(4)	(5)	6	(7)
a	8	(9)	10	(11)	(12)	(13)	14	(15)
	Gnd	a	Gnd	a	V_{cc}	V_{cc}	Gnd	V_{cc}

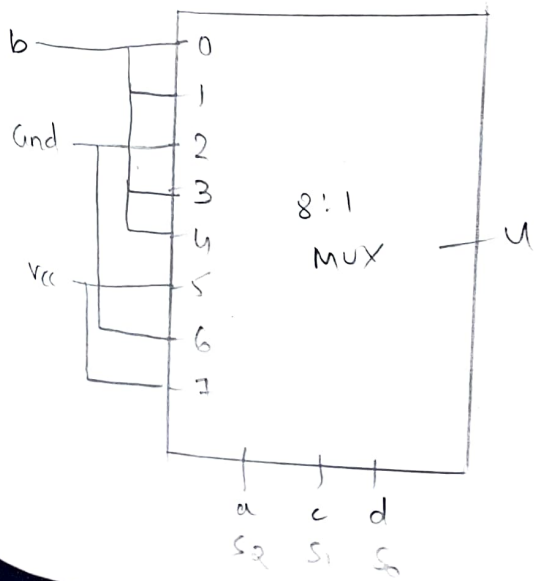
(3)





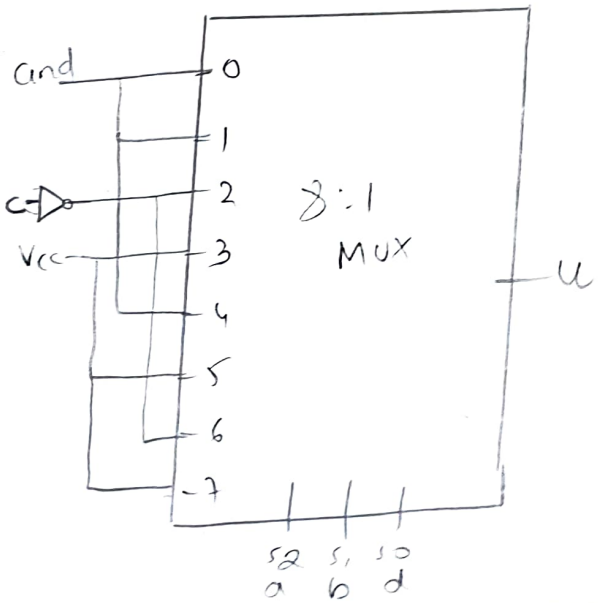
8:1 MUX → b to be i/p a c d select^n line

$\bar{b}$	0	1	2	3	8	(9)	10	(11)
b	(b)	(5)	6	(7)	(2)	(3)	14	(15)
	b	b	Gnd	b	b	V_{cc}	Gnd	V_{cc}



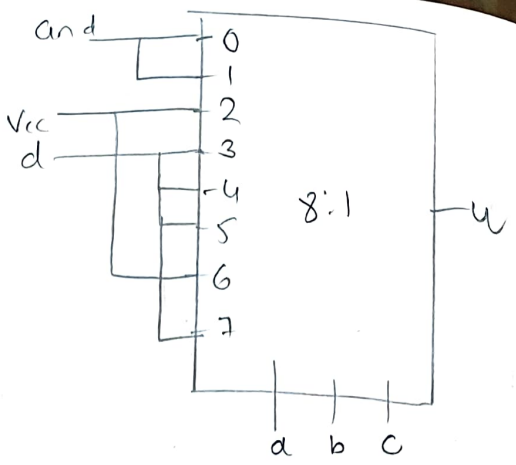
c as i/p a b d selection line

$\bar{c}$	0	1	(4)	(5)	8	(9)	(12)	(13)
c	2	3	6	(7)	10	(11)	14	(15)
	Gnd	Gnd	$\bar{c}$	V_{cc}	Gnd	V_{cc}	$\bar{c}$	V_{cc}



d as i/p and a, b, c as selection lines

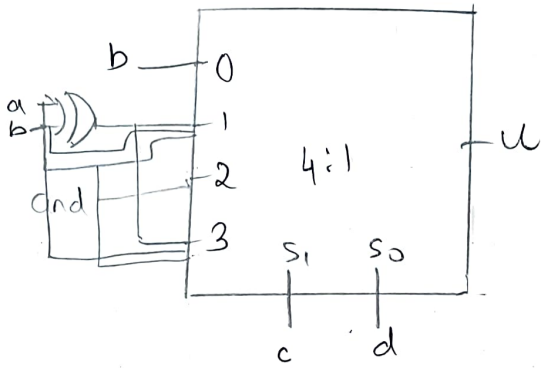
$\bar{d}$	0	2	(4)	6	8	10	(12)	14
d	1	3	(5)	(7)	(9)	(11)	(13)	(15)
	0	0	V_{cc}	d	d	d	V_{cc}	d



4:1

$\bar{a}\bar{b}$	0	1	2	3
$\bar{a}b$	(4)	(5)	6	(7)
$a\bar{b}$	8	(9)	10	(11)
ab	(12)	(13)	14	(15)
	b	$\bar{a}b + a\bar{b}$	and	$\bar{a}b + a\bar{b} + ab$

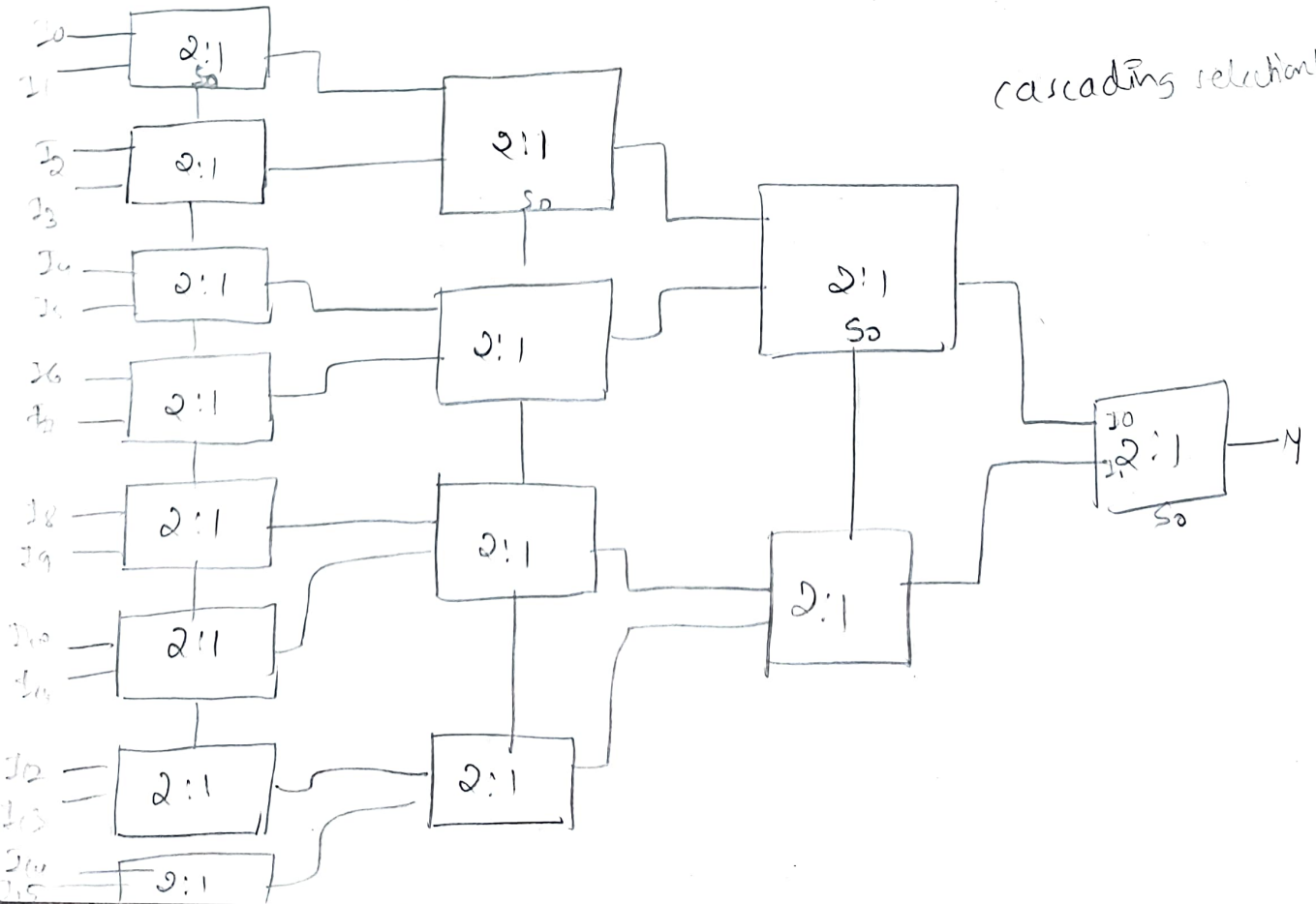
I_0 I_1 I_2 I_3



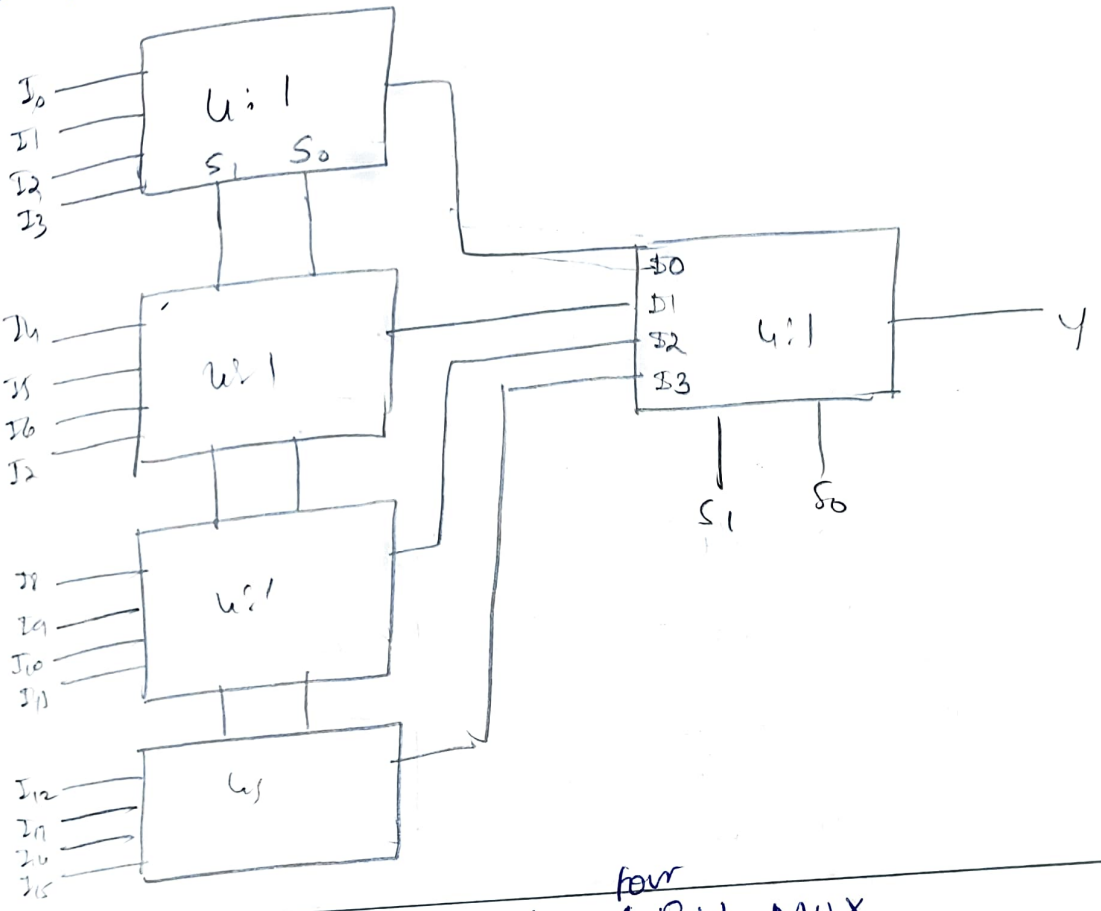
Implement 16:1 MUX using 2:1 MUX.

Enable always to gnd

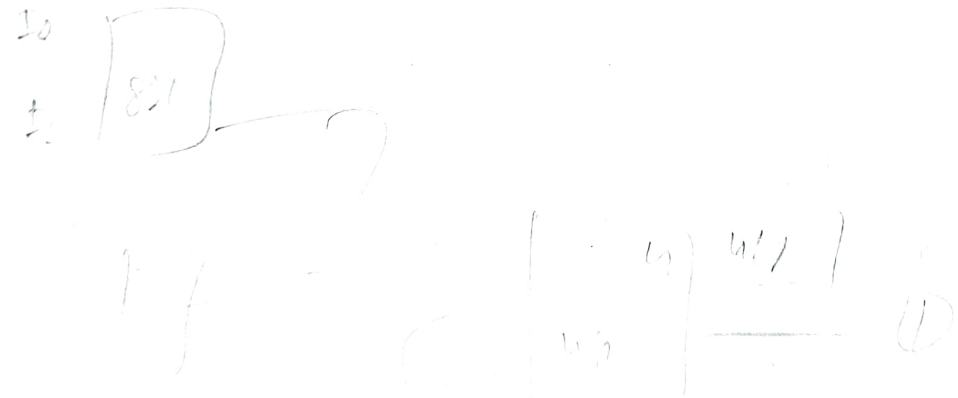
cascading selection lines



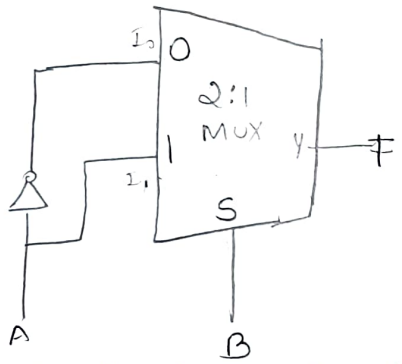
Implement 16:1 MUX using 4:1 MUX



Configure 13:1 MUX using four 8:1 MUX.



1) Consider the following circuit which uses a 2:1 MUX (GATE 98) as shown in the figure. Find the boolean expression for o/p y or F in terms of a & b.

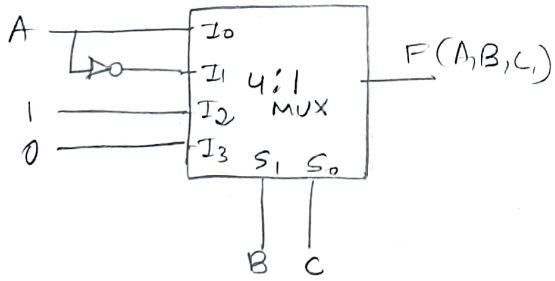


$F = \bar{S} I_0 + S I_1$

$F = \bar{B}\bar{A} + BA = \overline{A\bar{B}} + AB$

$y = \overline{A \oplus B} = A \odot B$

2) A 4:1 MUX is used to implement a 3 i/p boolean function as shown in fig. Find the boolean fn $F(A,B,C)$



$y = \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1\bar{S}_0 I_2 + S_1 S_0 I_3$

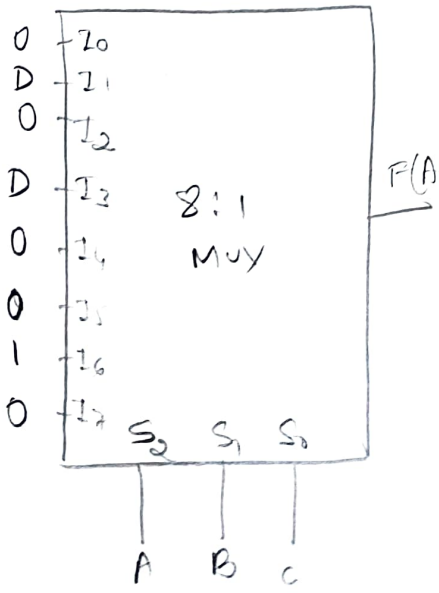
$= \bar{B}\bar{C}A + \bar{B}C\bar{A} + B\bar{C}\cdot 1 + BC\cdot(0)$

$= \bar{B}\bar{C}A + A\bar{B}\bar{C} + \bar{A}\bar{B}C + B\bar{C}(A+\bar{A})$

$= A\bar{B}\bar{C} + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$

$= \Sigma m(1,2,4,6)$

3) An 8:1 MUX is used to implement a logical fn y as shown fig. Find the o/p y in boolean expression.



$F = \bar{S}_2\bar{S}_1\bar{S}_0 I_0 + \bar{S}_2\bar{S}_1 S_0 I_1 + \bar{S}_2 S_1\bar{S}_0 I_2 + \bar{S}_2 S_1 S_0 I_3 + S_2\bar{S}_1\bar{S}_0 I_4 + S_2\bar{S}_1 S_0 I_5 + S_2 S_1\bar{S}_0 I_6 + S_2 S_1 S_0 I_7$

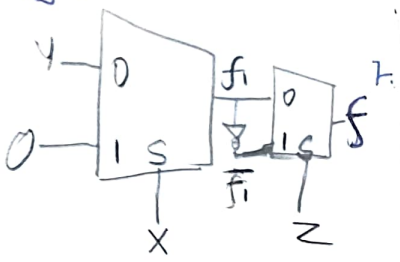
$= \bar{A}\bar{B}\bar{C}(0) + \bar{A}\bar{B}C(D) + \bar{A}B\bar{C}(0) + \bar{A}BCD + A\bar{B}\bar{C}(0) + A\bar{B}C(0) + AB\bar{C}(1) + ABC(0)$

$= \bar{A}\bar{B}CD + \bar{A}BCD + AB\bar{C}$

$= \bar{A}CD(\bar{B}+B) + AB\bar{C}$

$= \bar{A}CD + AB\bar{C}$

3) Consider the circuit shown below. Find boolean exp^n 'f'.



$f_1 = \bar{S} I_0 + S I_1$

$= \bar{X} Y + X(0)$

$= \bar{X} Y$

$\overline{\bar{X} Y} = \bar{\bar{X}} + \bar{Y}$

$= X + \bar{Y}$

$f = \bar{S} f_1 + S \bar{f}_1$

$= \bar{Z} \bar{X} Y + Z(\overline{\bar{X} Y})$

$= \bar{X} Y \bar{Z} + Z(X + \bar{Y}) = \bar{X} Y \bar{Z} + XZ + \bar{Y} Z$

4) Implement the f^n $f = A \odot B \oplus C$ using 8:1 MUX and 4:1 MUX.

$(A \odot B) \oplus C$

$[\bar{A}\bar{B} + AB] \oplus C$

$(\overline{\bar{A}\bar{B} + AB})C + (\bar{A}\bar{B} + AB)\bar{C}$

$(\bar{A}\bar{B} \cdot \overline{AB})C + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

$(A+B)(\bar{A}+\bar{B})C + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

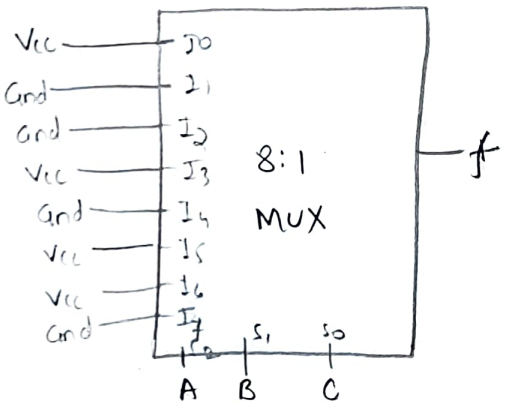
$(A+B)(\bar{A}C + \bar{B}C) + \;''$

$A\bar{A}C + A\bar{B}C + \bar{A}BC + B\bar{B}C + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

$A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

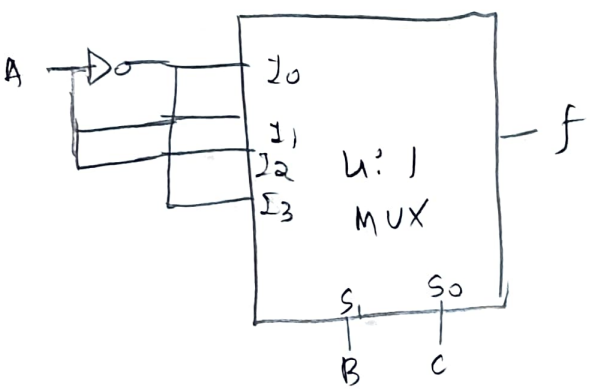
5 3 0 6

$\Sigma_m (0, 3, 5, 6)$



A	B	C	f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$\bar{A}$	⓪	1	2	③
A	4	⑤	⑥	7
	$\bar{A}$	A	A	$\bar{A}$
	I_0	I_1	I_2	I_3



Imp

Decoder:- combinational circuit ... n i/p and

gives 2^n o/p.

encoder → $2^n - n$

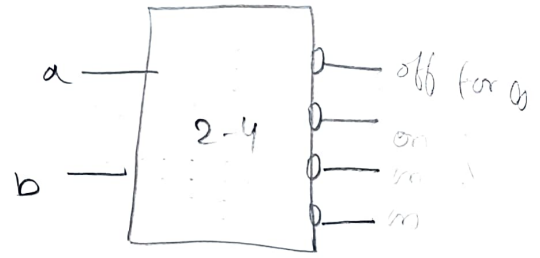
2-4 ✓ 74139

3-8 ✓ 74138

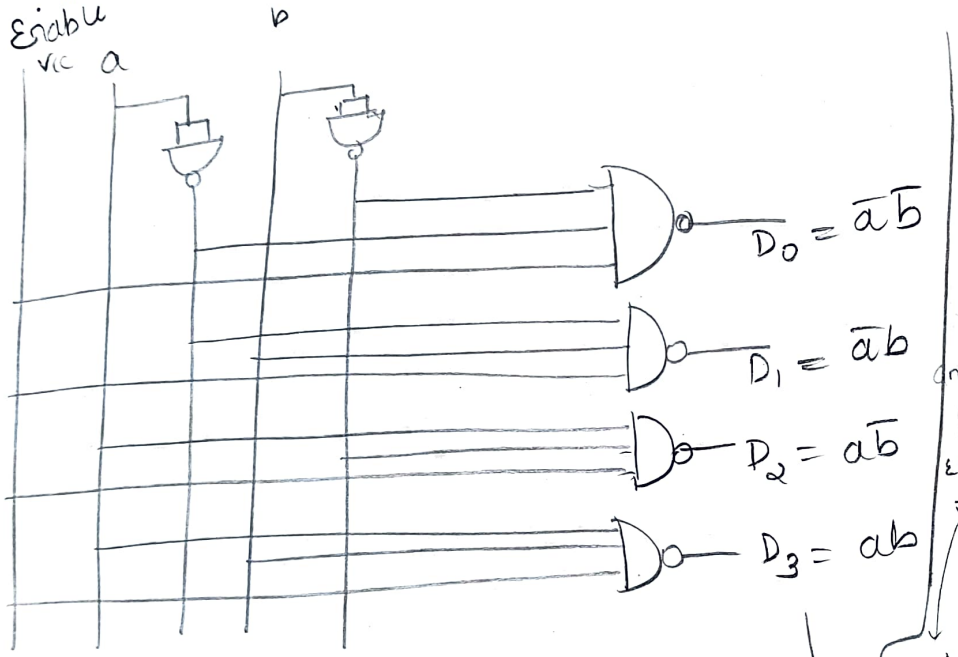
Active low o/p → AND gate

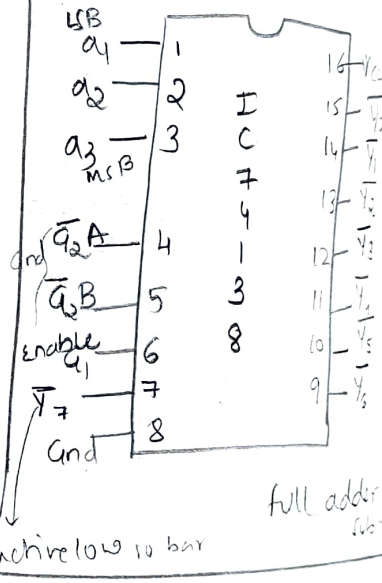
" high " → NAND "

Enable → high (1) → Vcc

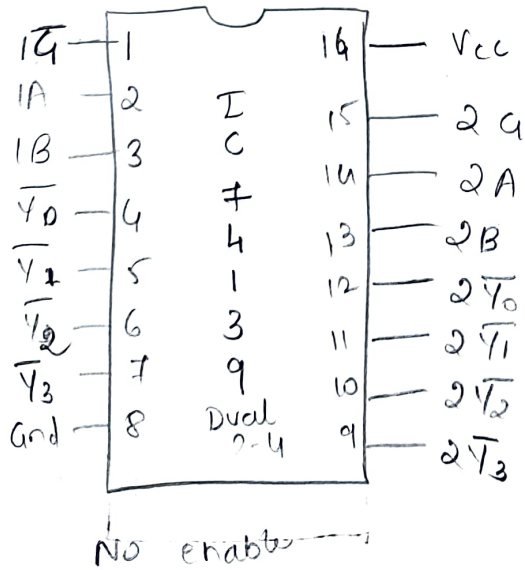


Implement 2-4 decoder using only NAND gates:





→ half adder & subb

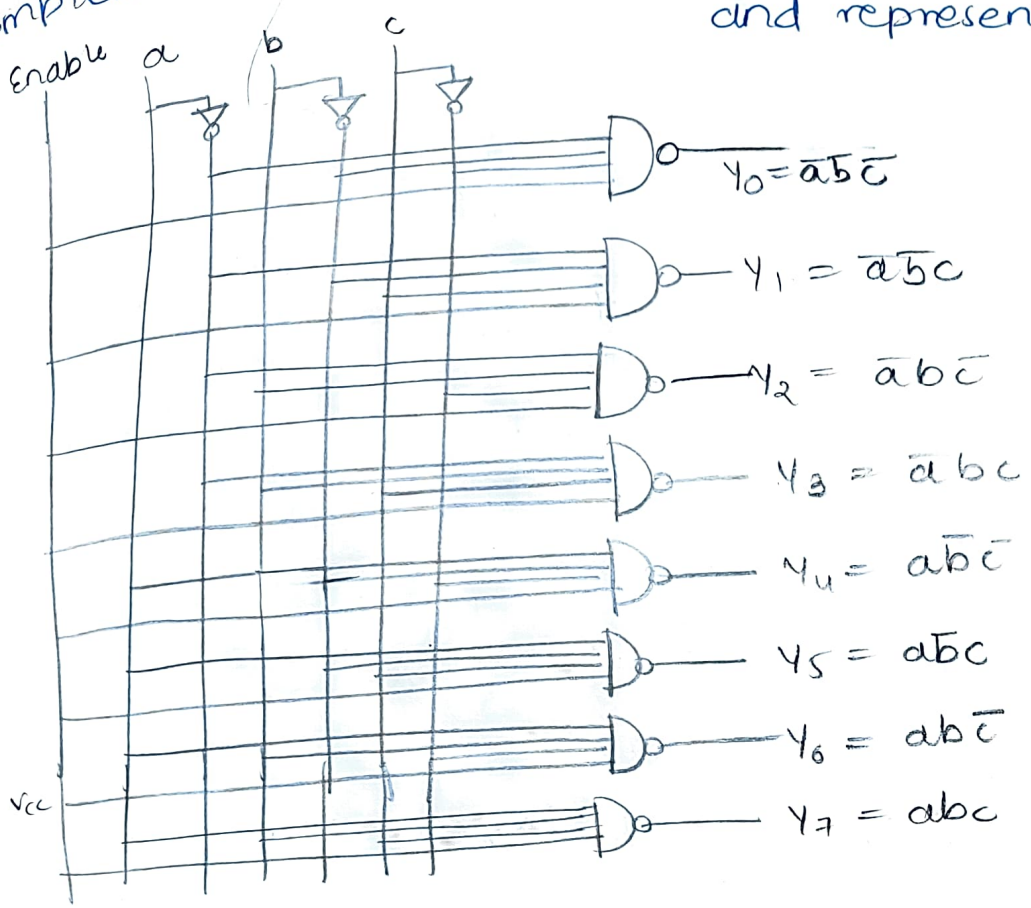


No enable

Truth table

I/p			O/p			
E	A	B	$\bar{Y}_0$	$\bar{Y}_1$	$\bar{Y}_2$	$\bar{Y}_3$
1	0	0	0	1	1	1
1	0	1	1	0	1	1
1	1	0	1	1	0	1
1	1	1	1	1	1	0

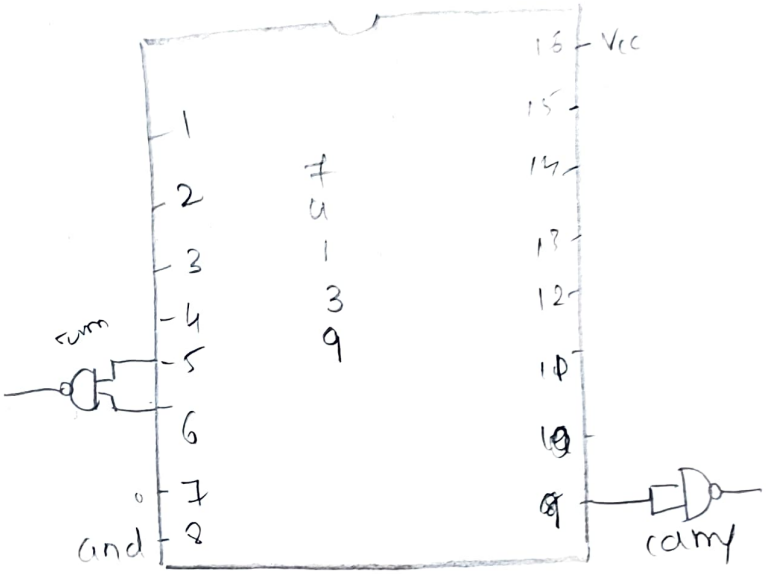
(2) Implement 3-8 decoder using ~~only~~ NOT and NAND gates. and represent truth table of the circuit.



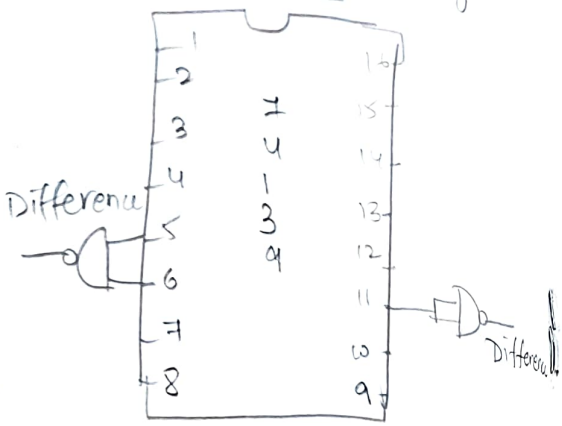
E	a	b	c	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
1	0	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	0	1	1	1	1	1	1
1	0	1	0	1	1	0	1	1	1	1	1
1	0	1	1	1	1	1	0	1	1	1	1
1	1	0	0	1	1	1	1	0	1	1	1
1	1	0	1	1	1	1	1	1	0	1	1
1	1	1	0	1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	1	0

(3) Implement half adder and half subtractor using dual 2-4 decoder (IC 74139).

Half adder



Half subtractor

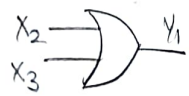


Encoders $2^n \rightarrow n$

4:2 encoder

X_0	X_1	X_2	X_3	Y_1 (MSB)	Y_0 (LSB)
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

$Y_1 = X_2 + X_3$



$Y_0 = X_1 + X_3$



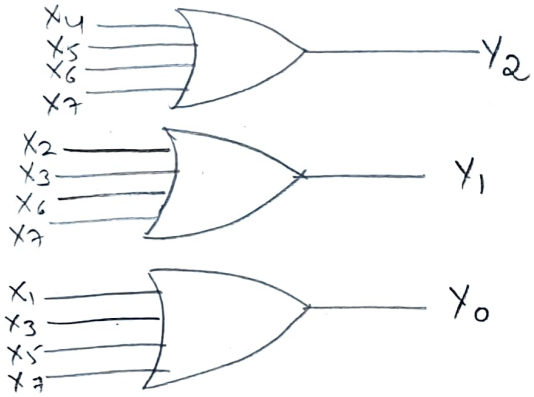
- Implement 8:3 encoder by representing truth table, boolean expression and logic diagram.

X_0	X_1	X_2	X_3	X_4	X_5	X_6	X_7	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$Y_2 = X_4 + X_5 + X_6 + X_7$

$Y_1 = X_2 + X_3 + X_6 + X_7$

$Y_0 = X_1 + X_3 + X_5 + X_7$



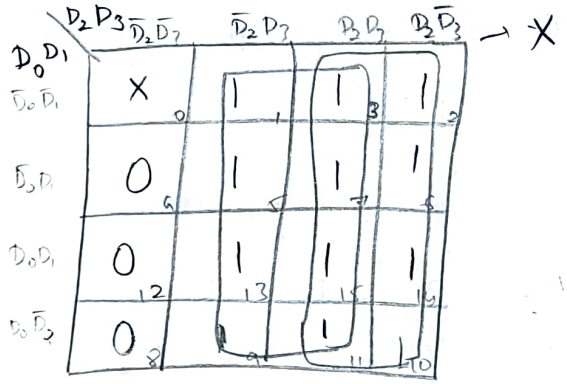
* Priority encoder

1) Design a priority encoder for the truth table shown below. Assume that D_3 has the highest priority and D_0 has the lowest priority.

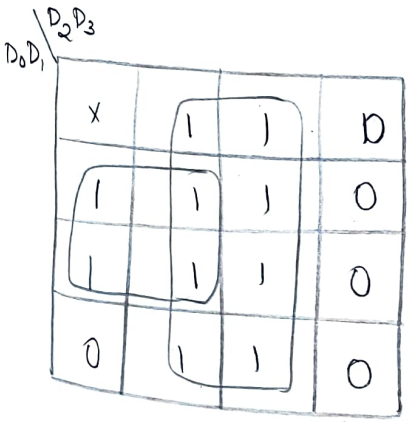
I/P D_0	D_1	D_2	D_3 (highest priority)	O/P X	Y	V (validity bit)
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

(lowest priority bit)

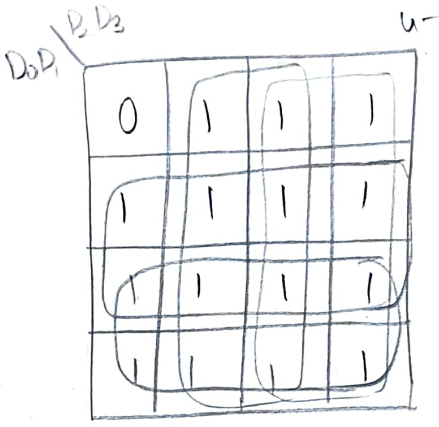
0 1 0 0



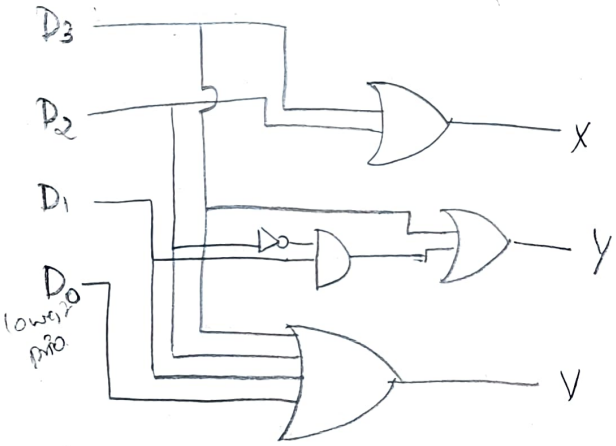
$X = D_3 + D_2$



$Y = D_3 + D_1\overline{D_2}$

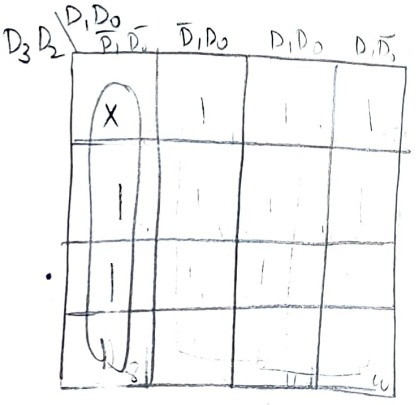


$$V = D_0 + D_1 + D_2 + D_3$$



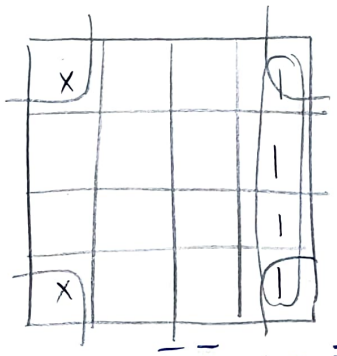
(2) Design a 4-i/p priority encoder with i/p and o/p described in the table below. Note that D_0 has highest priority and D_3 has the lowest priority.

D_3	D_2	D_1	D_0	X	Y	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

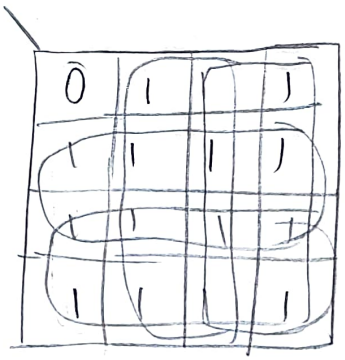


$$X = \bar{D}_1\bar{D}_0$$

0010
0011



$$Y = \bar{D}_2\bar{D}_0 + D_1\bar{D}_0$$



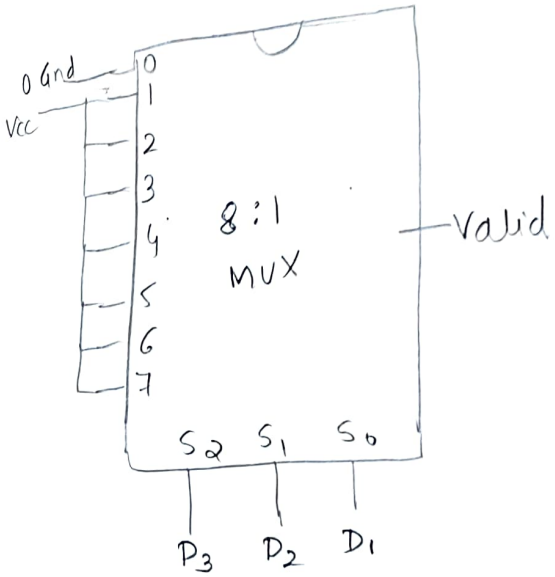
$$V = D_0 + D_1 + D_2 + D_3$$

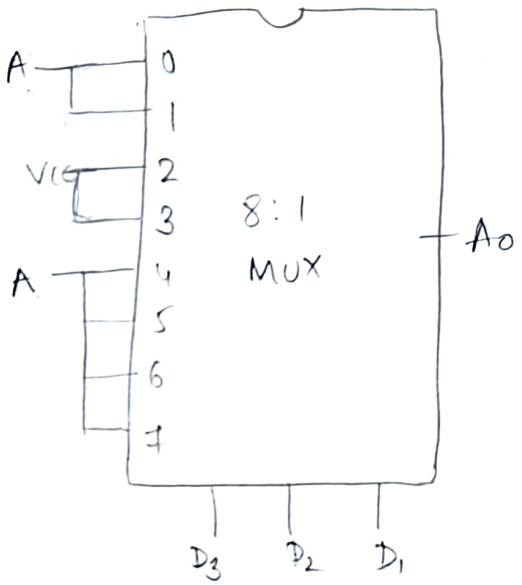
3) Implement 4:2 priority encoder using only 8:1 MUX and inverter.

Function is:

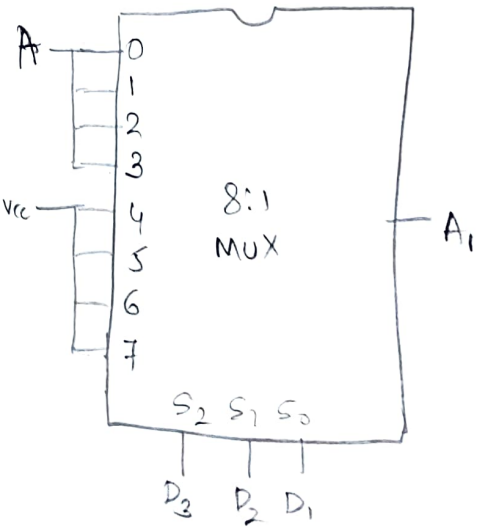
D_3	D_2	D_1	D_0	A_1	A_0	valid
0	0	0	0	x	x	0
0	0	0	1	0	0	1
0	0	0	x	0	1	1
0	0	1	x	1	0	1
0	1	x	x	1	1	1
1	x	x	x			

$\overline{A}$ 0 1 (2) (3) 4 5 6 7
A (8) (9) (10) (11) (12) (13) (14) (15)
M: 1 0 0 1 1 1 1 1





write truth table



$\overline{A}$ 0 1 2 3 (4) (5) (6) (7)
A (8) (9) (10) (11) (12) (13) (14) (15)
~~P3~~ ~~P2~~ ~~P1~~ ~~P0~~ Vcc Vcc Vcc Vcc

for A_1

If D_0 is taken as i/p

$\overline{D_0}$ 0 2 (4) (6) (8) (10) (12) (14)
D_0 1 3 (5) (7) (9) (11) (13) (15)
0 0 1 1 1 1 1 1

for A_0

$\overline{D_0}$ 0 (2) 4 6 (8) (10) 12 14
D_0 1 (3) 5 7 (9) (11) (13) (15)
0 1 0 0 1 1 1 1